Syllabus Year: 2024-2025

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - I

Course Code: BCA103-1 **Max Marks: 50**

Course Name: PRINCIPLES OF SOFTWARE DEVELOPMENT - I **Duration: 02:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Design and Develop ethical software product.

CO2: Apply terms, concepts, and tools of relational database management systems.

CO3: Construct database projects using DDL and DML queries.

Section A **2*5=10**

1)	Examine the concept of a relational model and analyze its application in creating a database for managing student records. Differentiate between entity sets and relationships within this context.	**2**	(CO2, L4)
2)	Examine and analyze the different layers involved in software development, explaining how each layer contributes to the overall functionality of an ethical software product.	**2**	(CO1, L4)
3)	Examine and analyze how ethical considerations can impact the quality of a software product during the requirement elicitation phase.	**2**	(CO1, L4)
4)	Analyze the concept of a relational model and differentiate between entity relationships and views in terms of their roles in database design.	**2**	(CO3, L4)
5)	Examine and analyze the importance of different layers in database management systems.	**2**	(CO2, L4)

Section B **5*4=20**

6)	Construct a relational database schema for a College Management System using DDL queries. Include at least three entity tables and one view that combines data from two of these tables. Provide examples of DML queries to insert, update, delete records in the main tables, and select records from the view.	**5**	(CO3, L3)
7)	Discuss the process of requirement elicitation and its quality assurance measures with an example.	**5**	(CO2, L2)
8) a)	Explain and demonstrate how the different layers of a software system interact with each other using an example from a typical web application stack.	**5**	(CO2, L3)

OR

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b)	Construct a scenario where you need to gather and analyze requirements for a new online retail management system. Demonstrate how potential issues during requirement elicitation can arise and explain your strategies to mitigate these issues.	**5**	(CO2, L3)
9)	Analyze the quality requirements that should be considered during the software development lifecycle and differentiate between their potential impacts at various stages of development.	**5**	(CO2, L4)

Section C **10*2=20**

10)	Explain and demonstrate how ethical considerations should be integrated during the requirement elicitation phase for an online education platform. Your answer should cover the following aspects: (i) Identify potential ethical issues that could arise during the development of such a platform, (ii) Discuss strategies to ensure these ethical issues are addressed in the requirements, and (iii) Describe how quality function deployment (QFD) can be used effectively in this context to align user needs with ethical standards.	**10**	(CO1, L3)
11) a)	Explain and discuss the significance of each layer in a multi-tiered application architecture. In your answer, illustrate how ethical considerations can be integrated into the design and development process at different layers.	**10**	(CO1, L2)
	OR		
b)	Explain the concept of a relational model in software development, focusing on its application in designing a database for a Library Management System (LMS). Additionally, discuss how ethical considerations can impact the design and development process of such a system. Your answer should be structured into three parts: (i) Describe the key components of the relational model; (ii) Illustrate the creation, insertion, and deletion of views in this context with examples; (iii) Analyze potential ethical issues that may arise during the implementation of the LMS and suggest ways to address them.	**10**	(CO1, L2)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create

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